MATH3033: Partial Differential Equations (Spring 2025)
Test 1

---

1. [30 marks] Solve the PDE $u_x + 2xy^2 u_y = 0$ with the auxiliary condition $u = y^2$ when $x = 0$

2. [30 marks] Use direct substitution to show that

$$u(x,t) = \frac{1}{2}[\phi(x+ct) + \phi(x-ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(s)ds$$

   solves the initial value problem $u_{tt} - c^2 u_{xx} = 0$ with the initial conditions $u(x,0) = \phi(x)$ and $u_t(x,0) = \psi(x)$

3. [40 marks] Use the Fourier transformation to show $u(x,t) = (g_t \star \phi)(x)$, where $\star$ denotes convolution and $g_t := \mathcal{N}(0, 2a^2 t)$ is heat kernel, solves

$$\begin{cases} u_t = a^2 u_{xx} \\ u(x,0) = \phi(x) \end{cases}$$

   [Hint]

$$\int \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} e^{-i\lambda x} dx = e^{-i\mu\lambda - \frac{1}{2}\sigma^2\lambda^2}$$